

ANAND KUMAR

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RESEARCH INTERESTS

Bilevel optimization; iterative algorithms and sparse reconstruction; machine learning; computational mechanics and structural analysis.

RESEARCH EXPERIENCE

Indian Institute of Technology Bombay Jan 2019–Present Graduate Research, Department of IEOR

- Research on two-timescale algorithms for bilevel optimization.
- Study of projection-based iterative algorithms for eigenvalue computation and Kaczmarz-based methods for sparse reconstruction.
- Seminar on the control of neutral-type time-delay systems arising in oil-well drilling vibrations.

Indian Institute of Technology Kharagpur Department of Civil Engineering

Multiaxial Fatigue Life Prediction May 2017–Apr 2018 Master's dissertation; Advisor: Prof. Arghya Deb

- Investigated extending the highly stressed volume (HSV) approach to multiaxial fatigue analysis using the finite element method (FEM).
- Validated the combined HSV–FEM approach for uniaxial fatigue, obtaining accurate life estimates; identified limitations in multiaxial predictions despite parameter modifications.

Stress Concentration in Nonaxisymmetric Rebars May 2016–Jul 2017 Bachelor's dissertation; Advisor:

Prof. Arghya Deb

- Investigated the increase in stress concentration beyond an optimal notch radius in rebars with asymmetric transverse ribs, using finite element simulations.
- Developed analytical models in MATLAB using numerical integration and solid mechanics to study stress components at a critical point.
- Identified the influence of shear stresses associated with cross-sectional area and shear-center variations on the maximum principal stress.

Structural Damage Detection Using Metaheuristics

May 2016–Jul 2016 Research internship project;

Advisor: Prof. D. Maity

- Implemented structural and eigenvalue analysis of three-dimensional frames in MATLAB, modeling damage through reductions in element elastic moduli.
- Used a frequency response function (FRF)-based objective with ant colony optimization (ACO) and particle swarm optimization (PSO) to identify damaged elements.

PUBLICATIONS

- Optimum rib design in TMT rebars to enhance fatigue life while retaining bond strength” (with Shrabani Majumdar, Riyaz Ali, and Arghya Deb). *Journal of Materials in Civil Engineering* (ASCE), 30(3), 04017313, 2018.

CONFERENCE PRESENTATIONS (ORAL)

- Well-Posed Reformulation of Bilevel Programming.” *5th International Conference on Nonlinear Applied Analysis and Optimization (ICNAAO 2025)*, Indian Institute of Technology Jodhpur, India, 4–6 Dec 2025.
- Lifting and Penalisation Approach for Mathematical Program with Vanishing Constraints.” *22nd Conference on Advances in Continuous Optimization (EUROPT 2025)*, University of Southampton, UK, 29 Jun–2 Jul 2025.
- Alternating Minimization for Least Square Estimation in Additive Semi-parametric Regression with Monotone Constraints.” *34th European Conference on Operational Research (EURO 2025)*, University of Leeds, UK, 22–25 Jun 2025.

WORKSHOPS AND SCIENTIFIC MEETINGS ATTENDED

- **4–15 Aug 2025:** *Data Science: Probabilistic and Optimization Methods II (DSPOM II)*, International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India.
- **23–25 Apr 2025:** *Workshop on Advances in Data-Driven Optimization*, Brij Disa Centre for Data Science and Artificial Intelligence, IIM Ahmedabad, India.

- **3–7 Jul 2023:** *Data Science: Probabilistic and Optimization Methods (DSPOM I)*, International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India.

INDUSTRY EXPERIENCE

Machine Learning Engineer at *Quantiphi Inc., India*

May 2018–Jan 2019

Worked on cross-category fashion apparel recommendation using deep neural networks, including convolutional autoencoders for style representations and similarity- and co-occurrence-based outfit recommendation.

Airframe Intern at *Airbus, India*

May 2016–Jul 2016

Worked with Rueben Dinakar on Python-based automation of explicit-to-implicit model conversion for large Abaqus models, examining differences between analysis procedures and applying object-oriented programming.

TEACHING EXPERIENCE

- **Teaching Assistant** at *IIT Bombay*: Courses: Optimization Techniques (IE 601); IEOR Lab (IE 684); Probabilistic Models (IE 502); Decision Analysis & Game Theory (IE 616, Jan–Apr 2023).
- **Lecture Series** at *National Institute of Oceanography, Goa* (Spring 2021): Scientific Programming with Python & Deep Learning, delivered as part of PMRF fellowship teaching requirements.
- **Teaching Assistant** at *IIT Kharagpur*: CAD & Engineering Drawing Laboratory (Autumn 2017); Undergraduate Mechanics (Spring 2018).

RESEARCH AND TECHNICAL SKILLS

Programming and tools: Python, MATLAB, Abaqus, $\text{\LaTeX}$, Linux.

Methods: Numerical optimization, finite element analysis, structural dynamics, eigenvalue analysis, sparse reconstruction, deep learning.

LEADERSHIP AND OUTREACH

- **Vice President**, Dakshana Alumni Network (2016).
- **Organizing Committee Member**, Megalith (2015).
- **Volunteer**, National Service Scheme (2014); weekly rural outreach for two years.